LLSMS2138 – Big Data in Finance

Instructions for group project 2024–2025

In groups of 4 or 5, address the questions provided below. Use your preferred software for that purpose (R, Matlab, Python, ...). I ask you to hand in a written report in pdf and a clear ready-to-work code. The project accounts for 20% or 30% of the final grade of the course (depending on which one gives you the best grade), based on the report and code only (no oral presentation or question at the exam).

Deadline: May 23

You can contact me at nathan.lassance@uclouvain.be to arrange a meeting in case you have questions on the project. Please do not come if you haven't seriously tried to solve the questions on your own beforehand.

The goal of the project is to apply different methodologies we have seen in the course to construct well-performing investment strategies on equity data. You will see that the questions are deliberately quite large to allow you to have some freedom and creativity in your answers. Please make sure that your report is easy to follow and not unnecessarily long. It is also important to display your results in a way that makes it easy to visualize the main conclusions. Finally, be precise: I need to see that you precisely understand what is going on and why.

To address the questions below, please use the following four datasets from Kenneth French's library (`mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html`), using the returns from January 1970 to December 2024: 25 portfolios formed on size and book-to-market, 100 portfolios formed on size and book-to-market, 10 industry portfolios, 48 industry portfolios. Take value-weighted returns and be careful that the returns are provided in percentage points, hence they should be divided by 100 before feeding them to your code. For each dataset, you will have to use both daily and monthly data. You are free to use additional datasets if you wish but, in that case, please join the data in a .txt file (in $T \times N$ format) along with your report and code.

Questions

1. Compare the in-sample and out-of-sample performance of three portfolio strategies: sample mean-variance portfolio, sample minimum-variance portfolio and equally weighted portfolio. For the in-sample performance, compute the three portfolios on all returns directly. For the out-of-sample performance, implement a rolling-window approach: use an estimation window of 10 years, compute the out-of-sample performance on the next six months, and then roll over the windows by six months until the end of the sample is achieved. Report the performance in terms of annualized mean, volatility and Sharpe ratio.

   Discuss your results and explain why you obtain such results. What is the impact of the return frequency (daily or monthly)? How stable are the portfolio weights over time (e.g. by computing the turnover or plotting boxplots)?

2. Following the same methodology than Question 1, report the out-of-sample performance of the ℓ_2-norm-constrained minimum-variance portfolio $\widehat{\boldsymbol{w}}_\gamma$ defined as

$$\widehat{\boldsymbol{w}}_\gamma = \underset{\boldsymbol{w}'\mathbf{1}=1}{\operatorname{argmin}}\ \boldsymbol{w}'\widehat{\boldsymbol{\Sigma}}\boldsymbol{w} + \gamma \boldsymbol{w}'\boldsymbol{w}.$$

   This portfolio has an analytical solution that you can directly use in your code, which is

$$\widehat{\boldsymbol{w}}_\gamma = \frac{\widetilde{\boldsymbol{\Sigma}}^{-1}\mathbf{1}}{\mathbf{1}'\widetilde{\boldsymbol{\Sigma}}^{-1}\mathbf{1}} \quad \text{with} \quad \widetilde{\boldsymbol{\Sigma}} = \widehat{\boldsymbol{\Sigma}} + \gamma \boldsymbol{I}_N.$$

   Why is adding a constraint on the ℓ_2 norm useful? How does the performance depend on γ? What do the portfolio weights look like compared to the sample minimum-variance portfolio?

3. Propose an optimal way of choosing the value of γ in Question 2. What is the resulting out-of-sample performance?

4. Repeat Question 3 for two other portfolio-weight constraints/penalties of your choice.

5. What is the out-of-sample performance of the mean-variance and minimum-variance portfolios when the covariance matrix is estimated via the first K principal components, as a function of K? Propose a way of choosing an optimal value of K and report the corresponding performance. Does Stein's eigenvalues estimate improve upon the sample eigenvalues?

6. Until now, you have ignored higher moments. To try to improve them, implement the IC-variance-parity portfolio using the FastICA algorithm. How do the first four moments (mean, volatility, skewness, excess kurtosis) and the modified Value-at-Risk of the portfolio return evolve as a function of K? What is the improvement in terms of skewness and kurtosis compared to the minimum-variance portfolio?

7. Implement an additional portfolio strategy coming either from a scientific paper of your choice or from the course. Please summarize the methodology and the objective of the strategy (e.g.

improve robustness, improve higher moments, reduce turnover, allow for big-data setting, ...). What is the out-of-sample performance of the strategy? **Important:** if you opt for a portfolio strategy from a scientific paper, please check the paper with me first.

8. What are your three most important conclusions following the analysis conducted above?

9. Suppose you are working in the investment industry and your manager asks you for a list of best practices in order to obtain a strong out-of-sample portfolio performance. What would you include in this list?

10. Most advertisements on investment companies include the usual warning "Past performance is no guarantee of future results." In view of your experience from the course and this group project, how would you adjust this warning to make it more accurate (in a single sentence)?

Question for a bonus point

You have the opportunity to gain one or half a bonus point if, relative to the other groups, you manage to choose a portfolio strategy that achieves the best out-of-sample performance on a dataset for which the out-of-sample data are kept secret from you.

On Moodle, you will find an excel file that consists of daily returns on $N = 300$ stocks from January 1980 to December 2009 (30 years of data). Your task is to find the portfolio strategy with the best possible Sharpe ratio on the next ten years; that is, from January 2010 to December 2019. However, these out-of-sample data are not made available to you.

Specifically, please follow those steps:

(1) Select a portfolio strategy of your choice and describe it at the end of your report.

(2) Along with your report, send me an excel file containing the vector of 300 portfolio weights computed via the method in (1).

I will then apply your portfolio weights in (2) on the out-of-sample data. The group that achieves the largest out-of-sample Sharpe ratio will gain one extra point, and the second best group half an extra point.